

Sean P. Florez

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Education

University of Colorado Boulder

Expected May 2028

Ph.D., Materials Science & Engineering. Advised by Prof. Hendrik Heinz in the Heinz Interfaces Laboratory.
Atomistic simulation of interfaces; experiment-calibrated force fields; autonomous-laboratory infrastructure.

University of Florida

May 2024

B.S., Materials Science & Engineering. Minor in Chemistry.

Research Experience

Graduate Research Assistant

University of Colorado Boulder, Oct 2024–Present

Heinz Interfaces Laboratory, Prof. Hendrik Heinz

Interlayer shear in MXenes (lead author, manuscript in preparation)

- Ran the production shear campaign for OH-, F-, and mixed-terminated Ti_3C_2 bilayers in LAMMPS with **780 NVT trajectories** across six termination and geometry cells on CU's Alpine cluster, ten seeds per stress, zero failed runs.
- Froze the protocol before production and had the campaign audited read-only afterward. The audit re-derived every reported critical stress from raw trajectories using an independent re-implementation of the slip classifier, and caught an inflated trajectory count in our own reporting before it reached the manuscript.
- Termination sets interlayer shear resistance across roughly a threefold range ($\tau_{0.5}$ 42–135 MPa), OH > mixed > F.

Hydrogen release from N-ethylcarbazole on Pt nanocrystals (lead author, manuscript in preparation)

- Designed and ran the MD campaign behind the manuscript, **60 slab production runs** across three Pt facets plus cuboctahedral nanoparticle ensembles, zero production job failures.
- Found that dehydrogenation-reactive configurations concentrate at low-coordination sites. Vertices carry $5.4\times$ the per-atom contact density of {100} terraces and $2.9\times$ that of {111}, so local coordination, rather than facet identity, sets where release happens.
- Built the per-area normalization methodology that makes facets, edges, and vertices comparable, and validated the finite nanoparticle against infinite slabs facet by facet.
- Wrote the technical narrative for the ALCF Director's Discretionary allocation that funds the next campaign.

Copper-oxide force fields for INTERFACE FF (industry-funded program)

- Parameterized INTERFACE force-field models for copper oxides as the first-year university deliverable, calibrated exclusively against experimental lattice constants, elastic constants, surface energies, and vibrational spectra.
- Delivered a validated Cu_2O parameter family in bonded 9-6 and 12-6 forms, anchored to experimental lattice constants and within $\pm 5\%$ of both independent experimental elastic references. Shipped as a distribution bundle (LAMMPS data files and run decks, an `.frc` for msi2lmp, a CHARMM/NAMD `.prm`), with all **52 configurations** run-verified.

Retrieval over the force-field corpus

- Built a SOAP-embedding retrieval index over the group's **3,009-structure, 6.7M-atom** corpus so a new atom's type and parameters can be proposed from its nearest structural neighbors behind a confidence gate. The same index maps coverage gaps and audits type consistency.

High-Performance Computing Intern

Air Force Research Laboratory, May–Aug 2025

Materials & Manufacturing Directorate, Wright-Patterson AFB

- Built the LAMMPS shear-testing workflow for OH-terminated Ti_3C_2 bilayers under controlled hydration and ran the stress sweeps on DoD HPC.
- Fit slip probability against applied shear stress with logistic models. Hydration turns out to be non-monotonic. Dry bilayers hold to 103 MPa, a quarter monolayer of water drops them to 8 MPa, and full coverage recovers to 39 MPa as water rebuilds bridging hydrogen bonds.
- First MD evidence that hydration tunes MXene interlayer shear strength by an order of magnitude, with the stress windows that follow for EM-shielding composite design.

Undergraduate Research Assistant

Hennig Group, University of Florida, Oct 2021–May 2024

- Benchmarked the VASPsol implicit solvation model against experimental energies from the Minnesota Solvation Database for neutral and charged solutes in water.
- Optimized cavity and screening parameters by grid search and Nelder–Mead across a large DFT campaign, and outlined a surrogate-model strategy that replaces high-count parameter sweeps for future calibrations.

Materials Modeling Intern

Air Force Research Laboratory, May–Dec 2023

- Built and validated an automated DFT workflow (Quantum ESPRESSO, ASE, and pymatgen) for high-throughput screening of low-work-function emitters, giving the group a reproducible selection path for field-emission candidates.

Research Automation & Compute Infrastructure

OpenSDL, computational and autonomous laboratory framework

2026–Present

github.com/fl-sean03/OpenSDL | seanflorez.com/OpenSDL (documentation) | Apache-2.0

- An open foundation for building computational and autonomous laboratories. Declare what a laboratory can do, execute those declarations as reproducible workflows, preserve the evidence, and feed the result back into the choice of the next experiment. Roughly 26,000 non-test lines of Python, about 600 test functions, and CI.
- A durable runtime with DAG execution, retries, timeouts, resource leases, restart reconciliation, and policy checks, over SQLite metadata and content-addressed artifact storage. Adapters for a simulated lab, local compute, and human tasks, with Slurm and hardware integration in progress; a CLI, Python SDK, HTTP API, and MCP hook; portable run export.
- Implemented campaign scoring and next-round selection. A reference three-dye formulation search matched an unseen target recipe within **0.5 percentage points**, evaluating **96 candidates per round**.

agent-fleet, persistent research automation

2026–Present

github.com/fl-sean03/agent-fleet

- Built a research harness based on a **15-agent fleet** operating for months, with isolated workspaces, persistent conversations, and inter-agent messaging.
- Ran multi-day computational campaigns with recovery after reboots, credential expiry, and preemption.

HPC & Cloud Throughput Infrastructure

2025–Present

github.com/fl-sean03/subjob | github.com/fl-sean03/cloudcomputemanager

- **subjob** runs many heterogeneous tasks inside one HPC allocation, so a hundred short jobs cost one queue wait instead of a hundred. Shared task pool, workers spanning nodes and partitions, per-task walltime enforcement, and heartbeat-driven recovery of a dead worker’s claims. Built for the Pt campaign’s per-snapshot analysis fan-outs.
- **CloudComputeManager** (MIT) manages GPU-cloud lifecycles for scientific workloads on spot instances, covering provisioning, environment setup, multi-stage pipelines, checkpoint and preemption recovery, batch sweeps, and cost tracking. Workload-agnostic, with 368 tests. Used for LAMMPS shear campaigns.

Manufacturing & Production Systems

Engineering Contractor, Rocky Mountain Scientific Laboratory

Jun–Aug 2026

Quality system for a client affiliate’s new energetic-materials production line
Automation & Robotics embed, Jun–Aug 2026

- Developed lot-release records covering inspections, approval authority, hold criteria, and nonconformance disposition, with item-level traceability to the governing **military specification**.
- Built a controlled, single-source document set with **10 client-format instructions**, **20 test-specific quality-control packets**, manufacturing procedures, and inspections.
- Automated checks for process-route isolation, link validity, terminology, and consistency between released documents and their source.
- Built a program tracker that generates the Kanban board, decision log, requirements traceability matrix, and design-review record.
- Supported quality, packaging, labeling, on-site commissioning, and hazard and failure-mode analysis.

National Security & Technology Transition

S&T Scouting Intern

The Joint Staff, J7 / Future Technology Office, Aug–Oct 2025

- Prioritized propulsion and energy concepts and produced **eight technical briefs** assessing readiness, risk, and experimentation paths for TRL 3–5 technologies.
- Coordinated with Service laboratories and DoD R&D organizations to connect emerging academic research to operational experimentation and transition pipelines.

Technology Commercialization Intern

Idaho National Laboratory (DOE), May–Aug 2024

- Ran **20 customer-discovery interviews** under Energy I-Corps and authored the commercialization plan for a laboratory-origin technology.
- Built AI and natural-language tooling for the technology-transfer team, and supported a DOE proposal that was awarded **\$2M**.

Leadership & Service

Co-Founder, [LabLink Initiative](#)

Aug 2024–Present

- Nonprofit moving community college students into national-laboratory research internships, principally the DOE Office of Science CCI and SULI programs. Partnerships with three Sacramento-area community colleges, and a webinar delivered across Phi Theta Kappa, the honor society for two-year colleges, that drew more than **250 students**.
- Built the matching pipeline and the tooling behind it.

Publications & Outputs

Co-author: *INTERFACE Force Field for Alumina with Validated Bulk Phases and a pH-Resolved Surface Model Database for Electrolyte and Organic Interfaces*. [arXiv:2601.12570](#), 2026 (preprint).

Lead-author manuscripts in preparation:

- *Modeling of MXene Shear Properties via Atomistic Molecular Dynamics Simulations*, with AFRL Materials & Manufacturing Directorate co-authors.
- *Understanding Hydrogen Release Mechanisms of N-Ethylcarbazole on Pt Nano-Crystal Catalysts*.

Open-source releases: [MolSAIC](#), a code-first MD system builder for atomistic model construction and force-field assignment; [pdb2msi](#), a PDB to Materials Studio structure converter.

Computing allocations: Argonne Leadership Computing Facility Director’s Discretionary award *HydrogenStorage*, approximately 26,150 node-hours across Polaris, Crux, Sophia, and Graphcore, awarded May 2026; CU Boulder Alpine and DoD HPC allocations for production MD campaigns, including 480-job Slurm arrays.

Quantum-chemical property prediction on QM9

[GitHub: qm9-smiles-predictor](#)

- Four reproducible pipelines (single-task, multitask, autoencoder, hybrid Mordred + ECFP4) predicting nine quantum-chemical properties across roughly 130k molecules.
- Reproduced and critiqued published SMILES-FNN baselines, showing where multitask learning underperforms and where hybrid descriptors reduce MAE over Mordred-only models, especially for dipole moment and polarizability.

Automated phase identification in experimental powder XRD

[GitHub: opxrd-ml-binary-phase](#)

- Ingested the roughly 92k-pattern opXRD archive (about 2.4% labeled) and formulated binary classification for a rare phase, with an interpolated 2θ grid and normalized intensities.
- Trained logistic-regression and random-forest classifiers with SMOTE for extreme class imbalance, reaching a 0.80 minority-phase F1 with balanced precision and recall on held-out data. A prototype for automated pXRD analysis in self-driving laboratories.

Technical Skills

Molecular dynamics: LAMMPS, NAMD, Materials Studio/Discover. Model building, equilibration protocol design, shear and deformation workflows, trajectory analysis at multi-GB scale.

Electronic structure: VASP, VASPsol, Quantum ESPRESSO. High-throughput workflows via ASE and pymatgen.

Force fields: INTERFACE FF, CVFF, CHARMM, PCFF, COMB3, EAM. Parameterization against experimental data, transferability testing, elastic and surface-energy validation.

Machine learning: scikit-learn, TensorFlow, imbalanced-learn, RDKit, Mordred, SOAP descriptors. Regression, classification, class-imbalance handling, surrogate modeling, uncertainty and error analysis.

Python: NumPy, SciPy, pandas, MDAnalysis, matplotlib. Package design, CLI tooling, pytest, ruff, pyright.

HPC and systems: Slurm and PBS Pro, job arrays, checkpoint-restart, throughput benchmarking, GPU and CPU-MPI builds; Linux, bash, git, systemd, SQLite, Docker, GitHub Actions, MCP.

Quality and manufacturing: Lot-release record and human-factors document design, military-specification traceability, requirements traceability matrices, PDR/CDR stage gates, FMEA and process hazard analysis, BOM development, vendor qualification, labeling compliance.

Security Clearance

Top Secret (active); SCI eligible.